

Differentiating AI-generated text from human-written text across LLMs

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Abstract

Distinguishing AI-generated text from human-written text is increasingly important for academic and professional contexts. We analyze 1,460 labeled samples from the Innovatiana dataset which consists of GPT-generated essays, supplemented with additional essays from multiple LLMs, using linguistic features to train Logistic Regression classifiers with Bag-of-Words and TF-IDF vectorizers as well as a DistilBERT classifier. Although both models achieve almost perfect accuracy on GPT-generated data, performance drops on previously unseen LLMs, revealing overfitting and limited generalization. Feature analysis shows human writing relies more on pronouns and everyday language, while AI text uses broader, formal vocabulary.

1 Introduction

The rapid advancement of generative artificial intelligence (AI) has made it increasingly important to develop reliable methods to distinguish AI-generated text from human-written text. As large language models (LLMs) become deeply integrated into education, workplaces, and digital media production, the ability to correctly identify human writing has significant implications for academic integrity, content authenticity, and the detection of AI-assisted plagiarism. Understanding whether a piece of text was written by a human or generated by an AI system is both a technical challenge and a broader ethical concern.

This study investigates the linguistic and stylistic features that differentiate AI-generated text from human-written text by training a set of classification models using the Innovatiana dataset of 1,460 samples. Although models trained on a single LLM can perform well, their generalizability to text produced by other systems remains an open question. To address this limitation, we supplement the original dataset with additional samples generated by

Claude, Grok, and Gemini, allowing us to evaluate the robustness across models.

By combining a sizable dataset, multiple models, and LLM generalization tests, this study contributes to the current literature that seeks to understand not only whether AI-generated text can be reliably detected, but also which linguistic signals drive this classification. Our findings aim to identify the strengths and limitations of current detection approaches and inform future methods to ensure transparency and accountability in AI-assisted communication.

2 Related Work

Prior research on distinguishing AI-generated text from human writing has generally focused on two objectives: improving classification accuracy and identifying the linguistic features that enable such distinctions. Some computational approaches rely primarily on handcrafted features. For example, [Reddy et al. \(2025\)](#) analyzed how perplexity, a measure of unpredictability, and burstiness, a measure of sentence and structure variety, inform the detection of AI-written text. While effective for a single model such as GPT, these approaches are limited in their generalizability across different LLMs. Other studies have explored automatically extracted linguistic features, including syntactic and lexical patterns, revealing consistent differences between human and AI writing, such as sentence complexity, vocabulary richness, and use of discourse markers ([Georgiou, 2025](#)). These findings complement handcrafted-feature approaches. Combining lexical, syntactic, and statistical signals may improve detection performance.

Human performance provides another point of comparison. Studies show that humans can distinguish AI-generated from human-written text with approximately 70% accuracy, with minimal differences across text domains ([Doru et al., 2025](#)).

This underscores both the difficulty of the task and the potential for computational methods to outperform humans. Traditional machine learning models, including Logistic Regression, SVM, and Naive Bayes, have been applied successfully, offering interpretable models that learn directly from data patterns rather than relying solely on hand-crafted rules. Building on these insights, our work extends prior research by combining a larger, more diverse dataset with multiple LLMs, allowing us to assess whether linguistic cues learned from one model generalize to others.

Overall, prior research provides essential groundwork but is often limited by corpus size or model-specific design. Our project follows a similar overarching goal of developing reliable methods to distinguish AI-generated from human-written text, while contributing a larger, more diverse dataset and explicitly testing generalization across multiple LLMs.

3 Methods

This study analyzes a corpus of 1,460 samples, consisting of both human-written and AI-generated text. The base dataset comes from the Innovatiana corpus, which includes essays across a range of topics. All text samples were labeled as either 0 (human-written) or 1 (AI-generated). 6% of the samples were AI-generated, which is likely designed to reflect real-world frequencies of human and AI writing in academia. Additionally, this encourages a more robust detection model because the small reference size makes AI-generated text harder to detect. The data was already cleaned and preprocessed.

To classify text samples as AI-generated or human-written, we trained two Logistic Regression classifiers, selected for their strong performance in prior work and its interpretability with respect to feature contributions. The dataset was split into an 80/20 train-test division, assigning higher weights to the minority class to address class imbalance. The first Logistic Regression classifier was trained using a Bag-of-Words vectorizer for the sake of simplicity. Next, we trained a Logistic Regression classifier using a TF-IDF vectorizer. TF-IDF is effective for this task because it highlights vocabulary differences by increasing the weight of words that appear frequently within one class but not across the whole corpus. This makes TF-IDF especially useful for text classification problems where lexical

patterns carry meaningful class information (Zeng, 2024).

Logistic Regression model performance was evaluated using accuracy, precision, recall, and F1-score. Confusion matrices were inspected to determine the types of misclassification errors most commonly made by the model. To identify linguistic properties that distinguish human from AI-generated, we extracted the top twenty words that were associated with both AI and human classification decisions. We then created side-by-side bar graphs to display the coefficients of each feature for the respective classes. We then plotted the reliability curve to assess model confidence and compare how the predicted probabilities align with the actual outcomes.

We trained the DistilBERT model using a weighted loss function to account for class imbalances, ensuring that both human and AI texts were appropriately represented. Training proceeded in epochs, with performance metrics (accuracy and loss) tracked at each epoch to monitor learning and convergence. After training, we evaluated overall accuracy as well as accuracy by category, providing a detailed view of how well the model performed on human versus AI-generated texts. Just as we had done for the Logistic Regressions, we conducted token-based feature analysis using gradient-based attribution, identifying the top twenty tokens most strongly associated with each class. These token importances were visualized in side-by-side bar plots to highlight distinctive lexical patterns. In addition, we computed a range of stylometric features for each essay, including sentence length, vocabulary richness, punctuation usage, and readability metrics, and presented these as comparisons in tabular format to assess broader structural differences between human and AI-writing.

One of the primary goals of this study was to evaluate how well different detection approaches generalized across language models. To address this, we tested both models on a small set of previously unseen AI-generated essays, with one essay produced by each of Claude, Grok, and Gemini. Each LLM was prompted to “write me an academic essay on any subject with no titles.” This allowed us to assess whether models trained on GPT texts could generalize to text produced by other LLMs and to identify potential limitations in robustness across models.

4 Results

4.1 Logistic Regressions

Using the Bag-of-Words vectorizer, the Logistic Regression classifier achieved strong classification performance, with an F1 score of 1.00 and a macro-averaged F1 score of 0.99. Similarly, the Logistic Regression model trained with a TF-IDF vectorizer performed well, achieving an F1 score of 0.99 and a macro-averaged F1 score of 0.97. These results indicate high overall effectiveness for both models on the test set and a potential case of overfitting.

Using the Bag-of-Words vectorizer, the model mostly latched onto raw frequency differences between the two classes. The AI-associated features were words like “its,” “state,” “and,” “votes,” “win,” “nationally,” “majority,” “exploration,” “decision,” “energy,” and “candidate.” These are pretty topic-heavy and lean toward political or generic informational writing. The human-associated words, on the other hand, were mostly function words and conversational markers, including “car,” “people,” “would,” “you,” “they,” “it,” “to,” “is,” “but,” and “on.” There’s a noticeable contrast in tone: the human texts are more personal, concrete, and narrative, while the AI texts sound broader and more formal. Because Bag-of-Words just counts word frequency, the model is capturing which kinds of topics each group tends to write about. However, these results should be taken with a grain of salt. The model’s reliability curve overshoots the diagonal at the mid-range, which may indicate overconfidence for probabilities that are not close to 0 or 1. The model is overall very confident and rarely outputs a middle value, and when it does, it’s unreliable there.

Using the TF-IDF vectorizer, the highest-weighted features for AI text shift toward more abstract and formal vocabulary, like “nationally,” “energy,” “trump,” “individual,” “exploration,” “ethical,” “ai,” “challenges,” “sustainable,” “technology,” “innovation,” and “diverse.” The human-text features under TF-IDF still include pronouns and basic connective words, like “the,” “to,” “car,” “cars,” “in,” “is,” “it,” “you,” “people,” “college,” “that,” “of,” “they,” “would,” “are,” and “we.” The fact that these remain distinctive even after TF-IDF reduces their weight suggests that human writers in this dataset rely much more on personal reference, everyday nouns, and casual language structures. However, the zig-zag pattern in the model’s reliability curve shows that the model’s mid-range probability

predictions do not match the actual frequency of AI texts. Even with the uneven shape, the curve still trends toward the ideal diagonal, which suggests the model generally assigns higher probabilities to texts that are more likely to be AI-generated.

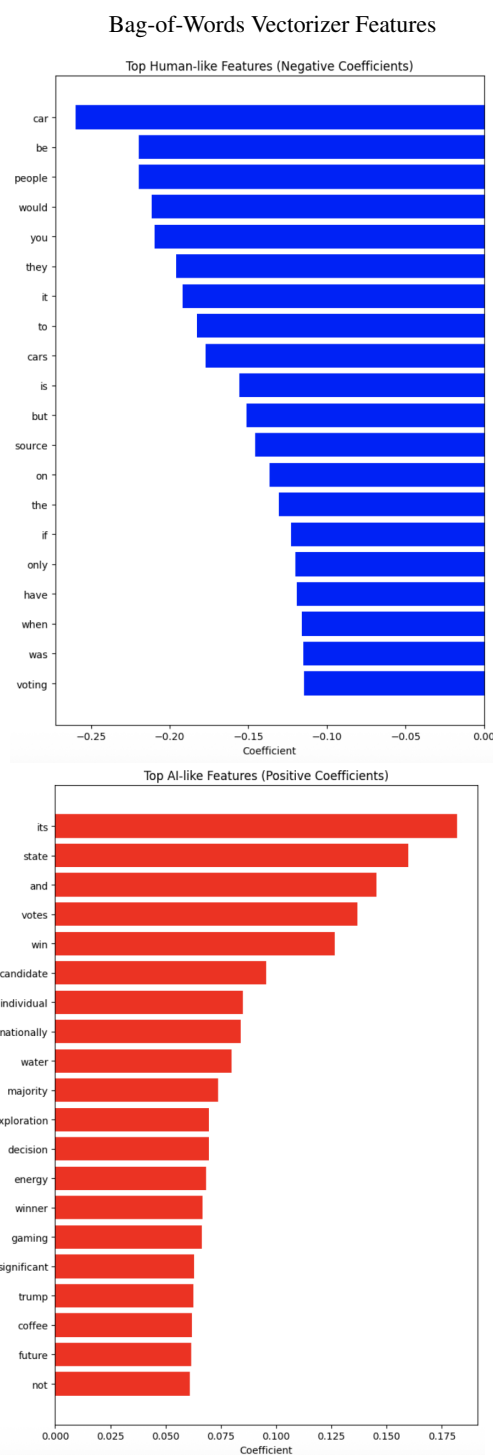


Figure 1: Top 20 tokens that inform human and AI classification for the Bag-of-Words vectorizer.

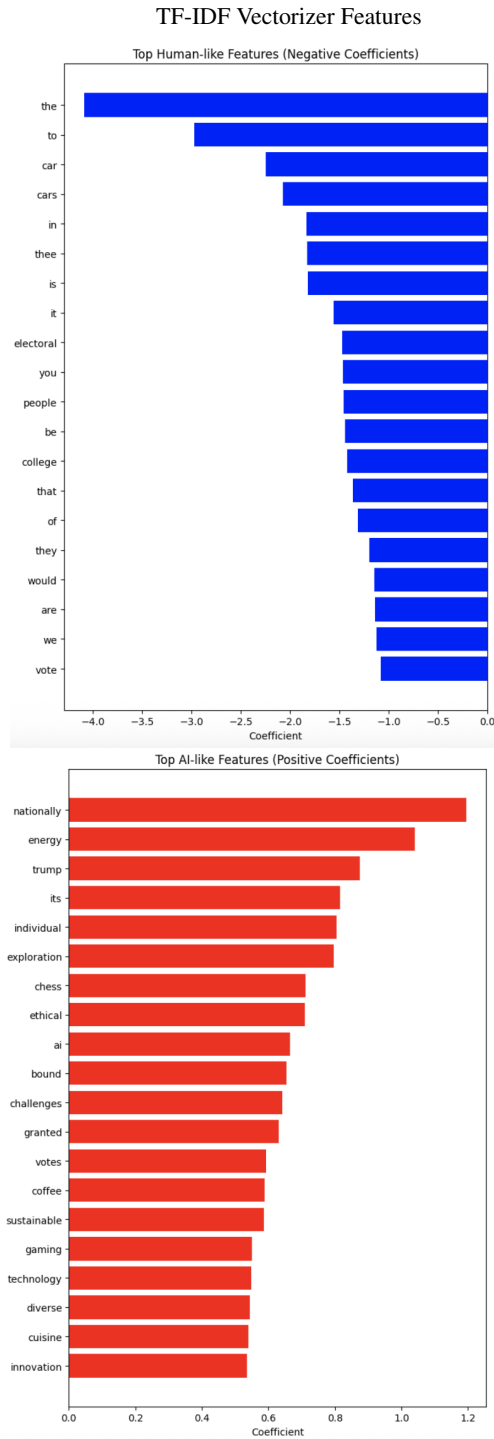


Figure 2: Top 20 tokens that inform human and AI classification for the TF-IDF vectorizer.

These models did not perform well on text generated by other LLMs. The model that used a Bag-of-Words vectorizer had an accuracy of 25%, and classified two probabilities below 1%, indicating strong confidence that the text was written by a human, despite it being written by AI. The TF-IDF vectorizer had an accuracy of 75%, and the essay that it misclassified had a probability of 49.8%.

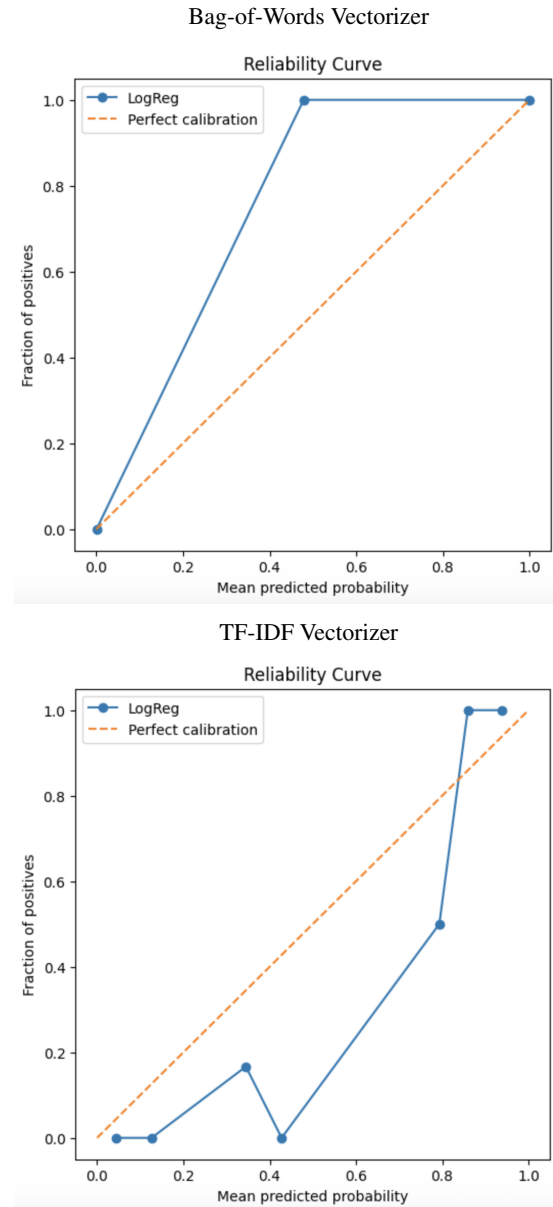


Figure 3: Reliability curves for both the Bag-of-Words and TF-IDF vectorizers, which show how model confidence compares to actual outcomes.

4.2 DistilBERT Model

The DistilBERT model achieved strong classification performance on the test set. Overall accuracy was 97.3%, with 99.6% for human-written texts and 95.0% for AI-generated texts. These results indicate that the model reliably distinguishes between human and AI writing, with slightly higher performance on human texts. The performance across epochs showed steady decreases in loss alongside increases in accuracy, suggesting stable convergence.

Examining the top tokens associated with each class provides insight into which words the model considers most indicative of human vs. AI writing.

For human texts, tokens such as “[CLS],” “the,” “states,” “have,” “you,” and “college” were most influential. These largely correspond to function words, personal references, and context-specific nouns. This highlights the model’s sensitivity to more narrative and concrete elements in human writing. AI-associated tokens included “state,” “and,” “senators,” “was,” “each,” and “total.” These tend to be more formal, topic-focused, or abstract, reflecting a broader and more informational writing style. The gradient-based attribution analysis confirms that the model emphasizes these distinctive lexical patterns when making predictions.

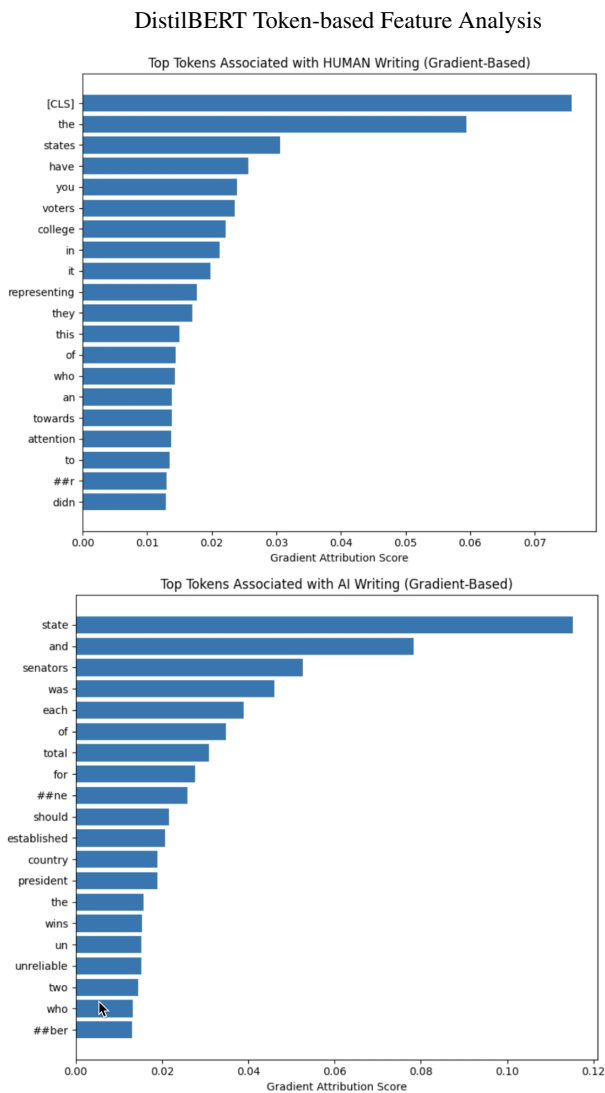


Figure 4: Top 20 tokens that inform human and AI classification for the DistilBERT model.

Stylometric comparisons between human and AI essays further corroborate these distinctions. Human-written texts exhibited a moderate sentence complexity, with an average length of 23.25 words

and a complex sentence ratio of 0.53. Vocabulary richness was lower (0.420) and rare word usage minimal (0.017), suggesting a more consistent, conversational style. In contrast, AI-generated texts showed higher syntactic complexity (complex sentence ratio of 0.84), slightly longer sentences (23.50 words), richer vocabulary (0.59), and more frequent use of rare words (0.116). AI essays also had lower Flesh Reading Ease (20.93), and higher Gunning Fog indices (19.08), indicating more formal and dense writing. Punctuation usage differed as well: AI texts contained more commas and dashes, whereas exclamation marks and quotations were more prevalent in human essays.

DistilBERT Stylometric Feature Analysis

Average Stylometric Features by Class

Feature	Human	AI
avg_sentence_length	23.2463	23.4995
sentence_length_var	122.1693	38.2315
complex_sentence_ratio	0.5335	0.8427
vocab_richness	0.4185	0.5872
rare_word_ratio	0.0172	0.1156
flesch_reading_ease	57.9583	20.9320
gunning_fog	12.7594	19.0760
comma_freq	0.0386	0.0701
semicolon_freq	0.0000	0.0003
exclamation_freq	0.0007	0.0000
quote_freq	0.0127	0.0050
dash_freq	0.0000	0.0107

Figure 5: A comparison of stylometric features including sentence-level features, word-level features, readability metrics, and punctuation/formatting metrics.

The DistilBERT model achieved 100% accuracy on the novel LLM dataset, correctly classifying all four essays from Claude, Gemini, Perplexity, and ChatGPT-5 as AI-generated. The model demonstrated high confidence across all predictions (probabilities ranging from 0.9937 to 0.9969). This is unlike the Logistic Regression models, which struggled significantly with generalization. DistilBERT’s superior performance on unseen LLMs suggests that its contextual architecture captures more generalizable linguistic patterns than traditional Bag-of-Words or TF-IDF approaches. It is also important to note that this evaluation is based on only four samples, and the high accuracy may not be representative of performance on a larger set of essays from these alternative models.

5 Conclusion

Our results demonstrate that machine learning models can distinguish AI-generated text from human-written text with very high accuracy under controlled conditions. When trained and evaluated

on GPT-generated data, both Logistic Regression models achieved near-perfect performance, substantially outperforming reported human accuracy in identifying AI-generated writing (Doru et al., 2025). However, this strong performance did not reliably transfer to text produced by unseen language models, revealing key limitations in generalizability when models are trained on a single AI source.

This limitation was especially pronounced for simpler lexical models. When evaluated on essays generated by other LLMs, the Bag-of-Words classifier achieved only 25% accuracy, while the TF-IDF model reached 75% accuracy, indicating partial but inconsistent robustness. In contrast, DistilBERT correctly classified unseen AI-generated essays with 100% accuracy, suggesting that transformer-based models better capture higher-level stylistic and structural cues rather than relying on GPT-specific lexical patterns.

Despite these differences in robustness, feature analyses revealed consistent linguistic trends across models. Logistic Regression associated human-written text with pronouns, basic connectives, and everyday vocabulary, reflecting a more personal and narrative style, while AI-generated text relied more heavily on abstract and formal language. DistilBERT's token-based analysis corroborated this distinction, showing that human writing is characterized by functional, context-specific tokens, whereas AI writing reflects more institutional and structured language patterns.

Stylometric analysis further reinforced these findings. Human-written essays exhibited greater sentence-length variability, while AI-generated texts showed more uniform sentence structures, higher syntactic complexity, and lower readability. Together, these results indicate that DistilBERT's strong performance arises from combining token-level and stylistic signals. At the same time, the sharp generalization failures observed in simpler models suggest that many detection systems still learn model-specific artifacts rather than universal features of AI-generated text, raising concerns about the long-term reliability of detection-based approaches as LLMs continue to evolve.

6 Limitations, & Ethical Considerations

This study has several limitations that constrain the generalizability of our findings. First, the dataset contains only 6% AI-generated samples, which,

while reflecting realistic academic proportions, limits the model's exposure to diverse AI writing styles and likely contributes to overfitting to GPT-specific patterns. In addition, our cross-model evaluation relied on only one essay per alternative LLM (Claude, Grok, Gemini, and Perplexity), a sample size insufficient for statistically robust conclusions about generalization. The reliability curves for both Logistic Regression models further suggest potential overconfidence, increasing the risk of unreliable predictions in ambiguous cases.

A second limitation is the assumption of binary authorship. Our models classify text as either human-written or AI-generated, but this framing fails to reflect the increasingly common reality of human-AI collaborative writing. Students frequently use AI tools for brainstorming, editing, or partial drafting, blurring the distinction between human and AI authorship. As AI becomes more integrated into academic workflows, binary detection systems may misrepresent how writing is actually produced and evaluated.

Finally, the ethical implications of misclassification are significant. False positive predictions, incorrectly labeling human-written text as AI-generated, can lead to serious consequences for students, including academic misconduct allegations, reputational damage, or even expulsion. These risks are asymmetric, as the harm caused by false positives often outweighs that of false negatives. Moreover, such systems may disproportionately impact non-native English speakers and students with highly structured or formal writing styles, which our models tend to associate with AI-generated text. This raises concerns about reinforcing linguistic and cultural biases and penalizing students for writing outside a narrow normative standard.

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